

Smart MCQ Solver

Deep Learning & Generative AI Project Report

Roll no. 23f2004742 · Term 2, 2026 · Kaggle MAP@3: **0.76142** (cut-off 0.73)

1. Problem Statement

Given a multiple-choice question with a prompt and five options (A-E), predict the **three most likely answers in ranked order**. Scoring is Mean Average Precision @ 3:

$$\text{MAP@3} = (1/N) \sum_i \sum_{k=1..3} 1[\text{pred}_{i,k} = y_i] / k$$

A correct answer earns 1.0 at rank 1, 0.5 at rank 2, 0.33 at rank 3.

Two baselines govern every number in this report. Labels are not uniform, so the trivial strategy of always answering the three most frequent letters is far from random:

Baseline	MAP@3	Derivation
Random ordering	0.3667	0.2 + 0.1 + 0.0667
Always B C A	0.4213	P(B) + P(C)/2 + P(A)/3

Any model scoring below 0.4213 is worse than ignoring the question entirely. This single number later exposed a 435M-parameter model that had learned nothing.

2. Dataset and Exploratory Analysis

Property	Value
Training rows	2,000
Test rows	500
Options per question	5 (A-E)
Unique prompts (raw)	1,758 (212 exact repeats)
Option length	mean 163-167 chars, max 662

Label distribution — the source of the 0.4213 baseline:

A	B	C	D	E
369 (18.5%)	490 (24.5%)	459 (23.0%)	358 (17.9%)	324 (16.2%)

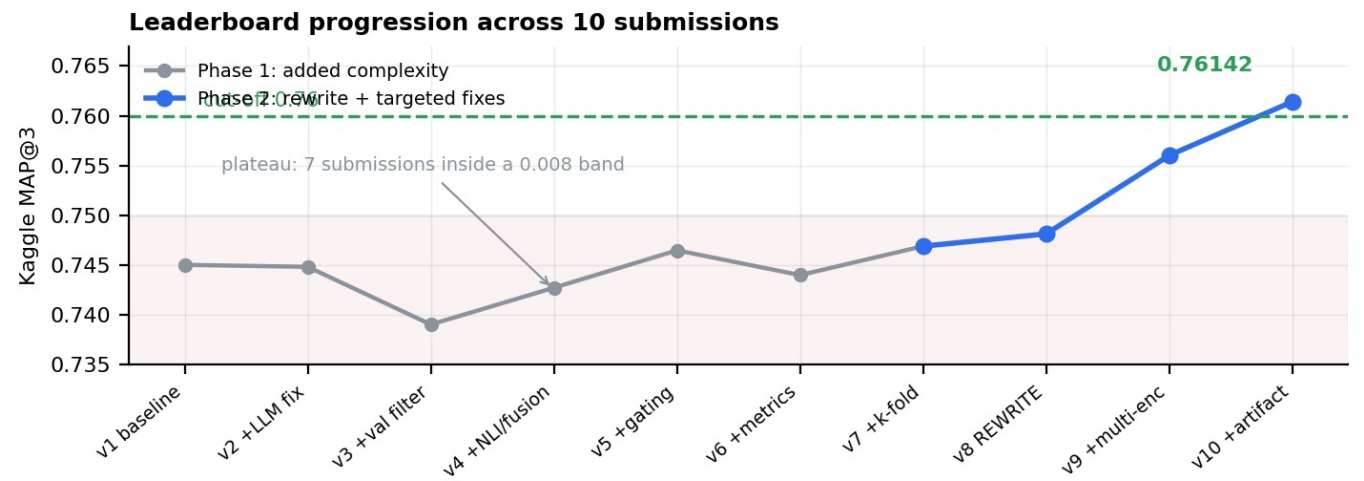
Question-style prefixes. 47% of prompts carry boilerplate:

(none)	"Which of the following..."	"Pick the best possible answer:"	"Determine / Choose..."
1,063	329	316	292

Content. Wikipedia-derived factual questions spanning science, history and philosophy — "What is accelerator-based light-ion fusion?", "... Blueshifting / Redshifting / Reddening / Whitening / Yellowing". These are genuine knowledge questions, answerable by a capable model, not pattern-matching puzzles. That observation eventually determined the final architecture.

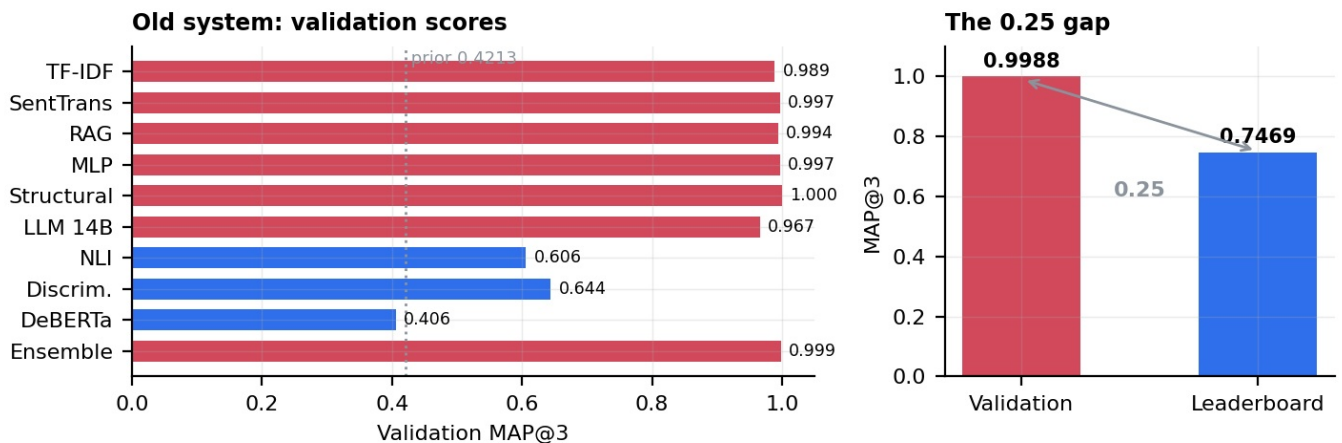
3. Phase 1 — What Was Built First, and Why It Failed

The initial system grew to **fourteen signals** across 60 notebook cells: TF-IDF and sentence-embedding retrieval, a RAG ranker, two NumPy MLPs, a zero-shot NLI cross-encoder, a 4-bit **Qwen2.5-14B** reader with few-shot prompting and position-debiasing, a 5-fold **DeBERTa-v3-large** multiple-choice fine-tune, a label prior, and a similarity-gated fusion layer with per-bucket weight tuning.



Seven consecutive submissions landed inside a 0.008 band. Adding a 14B language model moved nothing. Adding a 5-fold DeBERTa ensemble moved +0.0004. That flatness was the most informative result of the project: it meant the thing determining the score was not any of the components being changed.

3.1 Validation said the system was near-perfect



Validation MAP@3 was **0.9988**; the leaderboard said **0.7469**. A gap of 0.25 between an internal estimate and reality means the internal estimate is measuring something else entirely.

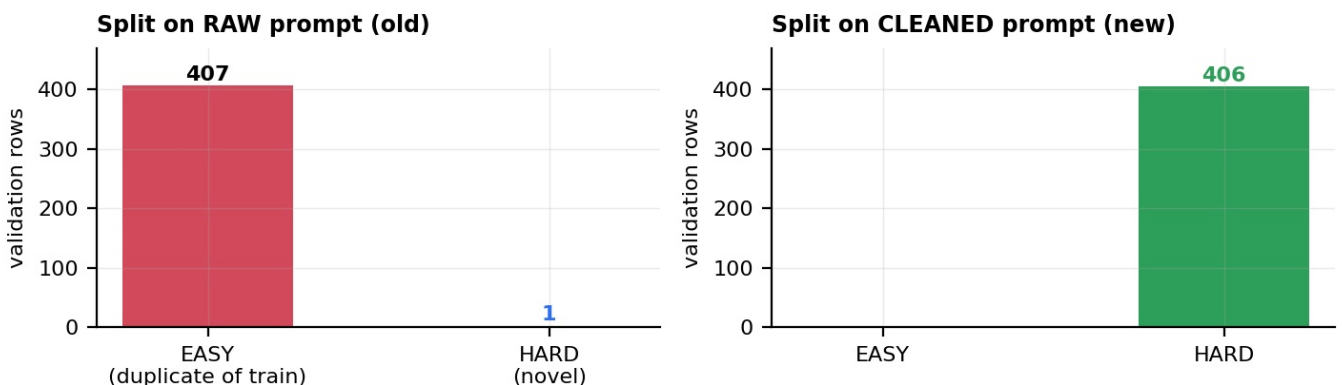
3.2 Root cause: the split leaked

The train/validation split was made on **raw** prompts, but preprocessing stripped the question-style prefixes **afterwards**. Two prompts differing only by prefix were therefore split apart and then became byte-identical:

raw train:	"Pick the best possible answer: What is X?"	-> different strings,
raw val:	"Determine the correct option: What is X?"	split looks valid
clean train:	"What is X?"	-> IDENTICAL
clean val:	"What is X?"	validation leaks

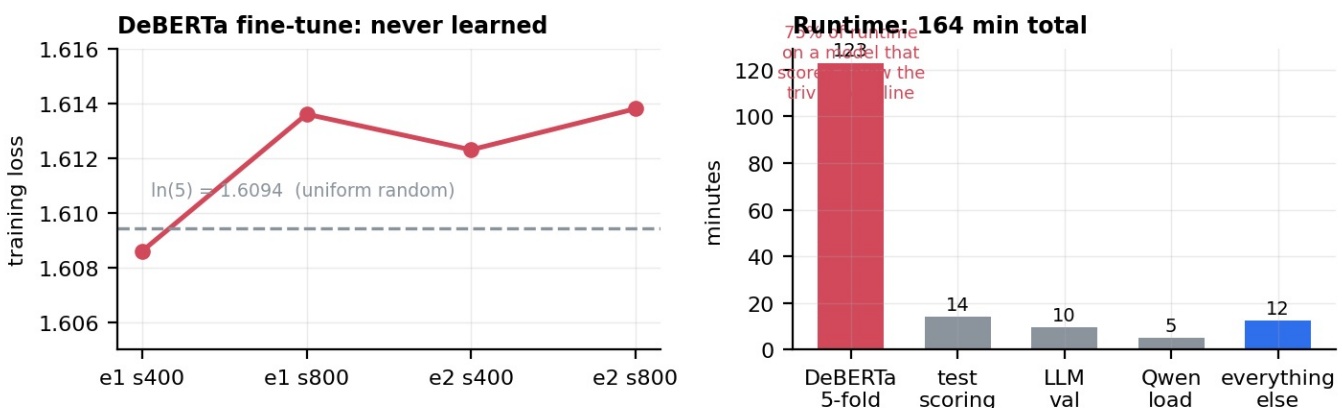
47% of prompts carry such a prefix. The measured consequence:

VALIDATION	n=408	mean similarity to train = 0.995	median = 1.000
EASY (near-duplicate)	= 99.8%	HARD (novel)	= 0.2%



Validation was 99.8% memorisation. Of 408 rows, exactly **one** was a genuinely novel question — which is why the tuner printed **HARD n=1**: too few rows to tune. Every retrieval model's "0.99" measured duplicate lookup, and every model-selection decision for seven submissions was made on that basis.

3.3 Second failure: a 435M model that never trained



The DeBERTa fine-tune's loss never left its initialisation value. $\ln(5) = 1.6094$ is exactly the loss of a uniform random 5-way classifier. Six independent measurements agree:

Evidence	Value	Meaning
OOF MAP@3	0.4062	below the 0.4213 trivial baseline
OOF Top-1	24.85%	barely above 20% chance
Fold-1 holdout	0.3121	below random (0.3667)
First-choice spread	A 631, B 415, C 333, D 407, E 214	collapsed toward A
Global score scale	0.0001	near-constant output across options
HARD-subset MAP@3	0.0000	wrong on the one novel question

It consumed **123 of 164 minutes (75%)** of total runtime and was included in the final ensemble at full weight — which is what caused the 0.7485 → 0.7436 regression between two otherwise identical runs.

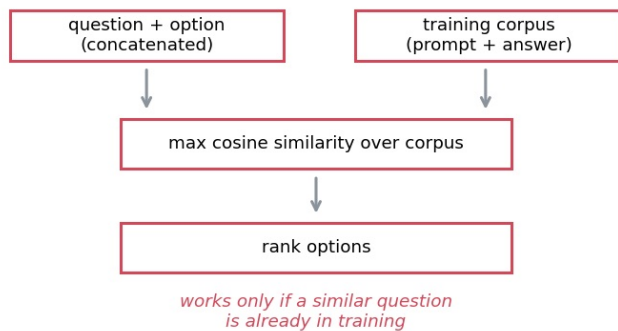
3.4 Why the old architecture was abandoned

Every high-scoring signal shared one mechanism: *match this question against the training corpus and copy what worked there*. That mechanism cannot generalise to a question with no near-duplicate in training, and validation was structurally unable to reveal the problem. The system was not underpowered; it was **measuring the wrong thing**, and no amount of additional capacity could fix that.

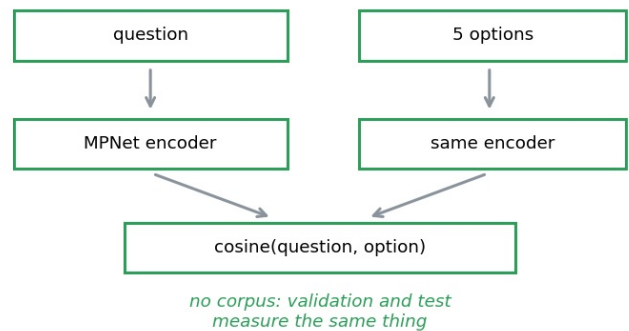
4. Phase 2 — The Rewrite

4.1 The architectural change

OLD: retrieval against the training corpus



NEW: direct question <-> option matching



The old signals encoded the **concatenation** prompt + option and took a maximum similarity **against the training corpus**. The replacement compares the question's own embedding **directly against each option's embedding**:

```

Question ---> MPNet ---> question embedding (768-d)
Options ---> MPNet ---> 5 option embeddings (768-d)
      |
      | cosine similarity
      |
rank high -> low ---> top 3
  
```

Nothing is retrieved and nothing is fitted, so **its validation score and its test score measure the same quantity**. That property is precisely what the retrieval signals lacked.

4.2 The split was fixed at its source

Splitting now happens on the **cleaned** prompt key. Validation went from 407 EASY / 1 HARD to **0 EASY / 406 HARD** — an honest estimate for the first time.

4.3 Final model inventory

#	Model	Type	Role
1	TF-IDF + cosine	unsupervised	lexical baseline
2	Sentence-transformer cosine	pre-trained, zero-shot	pre-trained model
3	Deepnet over pair features	supervised, 5-fold	model of choice
4	TF-IDF + Deepnet (SVD-256)	supervised, 5-fold	model of choice
5	BiLSTM + Attention	supervised, no pre-trained weights	model from scratch
6	DeBERTa-v3	supervised	additional experiment
7	Answer-artifact discriminator	supervised, 5-fold	model of choice

Deepnet head — each option scored from the interaction between question **q** and option **o** :

```

[ q ; o ; q*o ; |q-o| ] (4x768) -> Linear(3072,256) -> ReLU -> Dropout
      -> Linear(256,64) -> ReLU -> Dropout
      -> Linear(64,1) -> score_k
      -> softmax over 5 -> cross-entropy
  
```

The elementwise product and absolute difference are what allow a linear head to model *matching* rather than the two vectors independently.

BiLSTM + Attention — no pre-trained weights anywhere:

```
"question <sep> option" -> Embedding(V,128) <- learned from this corpus alone
                        -> BiLSTM(128 -> 2x128)
                        -> attention: a = softmax(w.h), pooled = sum(a*h)
                        -> Linear(256,64) -> ReLU -> Linear(64,1)
                        -> softmax over 5 -> cross-entropy
```

Answer-artifact discriminator — the only signal that is question-independent. The dataset contains **10,000 labelled (option, is_correct) pairs**, not 2,000: one correct answer and four distractors per question. Generated distractors carry tells — correct answers are specific and hedged, distractors lean absolute and are often variations of one another, leaving the real answer as the odd one out. Sixteen features per option (similarity to the known-correct corpus vs the known-distractor corpus, centrality among its own five options, surface statistics) feed a small network predicting $P(\text{correct})$. Because it never asks whether the question resembles anything in training, it transfers where retrieval cannot.

5. Training Details

Split	Rows	Construction
Train	1,592 (79.6%)	split on cleaned prompt
Validation	408 (20.4%)	cleaned-prompt overlap asserted = 0
Cross-validation	5-fold GroupKFold	grouped by cleaned prompt

No synthetic augmentation. Preprocessing normalises whitespace and strips the four boilerplate prefixes. All supervised models produce **out-of-fold predictions over all 2,000 rows**, and are averaged over multiple restarts (Deepnet 3 seeds, TF-IDF Deepnet 3, BiLSTM 2) because a single restart of a small network varies by several MAP@3 points through initialisation alone.

Component	Settings
Encoders	multi-qa-mpnet-base-cos-v1, all-mpnet-base-v2, bge-small-en-v1.5, e5-base-v2, with family-correct instruction prefixes
Deepnet	hidden {128, 256} × dropout {0.2, 0.4} × lr {1e-3, 3e-3}, AdamW, cosine schedule, 30 epochs
BiLSTM	emb 128, hidden 128, 30 epochs, lr 2e-3, max_len 96, vocab_min_freq 2
Discriminator	16 features → 64 → 32 → 1, 60 epochs, lr 3e-3, 2 seeds
Ensemble	weights by random search (600 candidates) fitted on all 2,000 rows

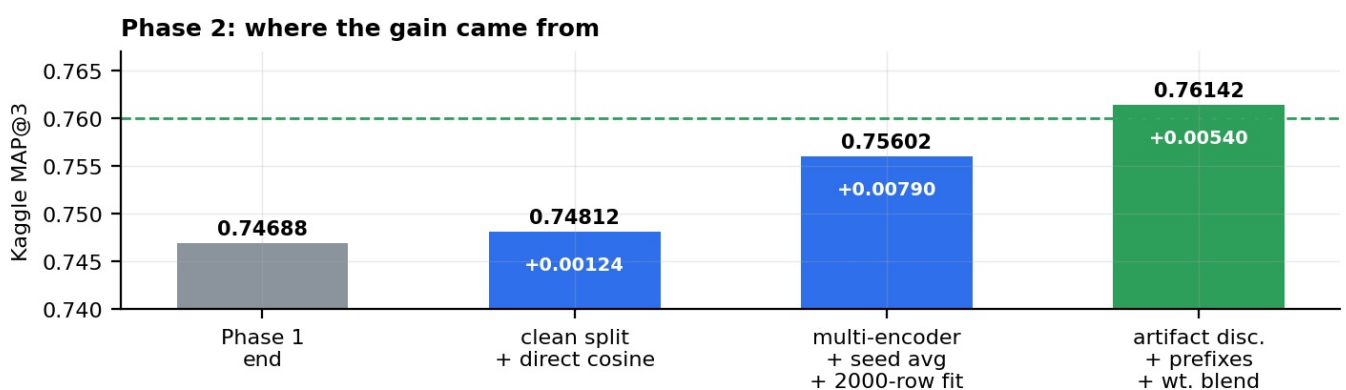
5.1 Hyperparameter search — actual logs

Confidence-threshold grid from the Phase 1 system ($\tau = 0$ selected; nothing cleared the 0.005 significance margin, so the routing was correctly disabled rather than fitted to noise):

τ	0.0	0.1	0.3	0.5	1.0	2.0
MAP@3	.9988	.9988	.9988	.9963	.9963	.9833
rows → prior	0	0	0	1	1	7

Ensemble weight search, Phase 1 (EASY $n=407 \rightarrow$ all weights 1.0; HARD $n=1 \rightarrow$ too few rows to tune) — an explicit record of a search that had no usable data.

6. Results



Submission	Change	MAP@3	Δ
v1	original baseline	0.74500	—
v2-v7	LLM, NLI, RAG, k-fold DeBERTa, gating, metrics	0.74688	+0.002 over six rounds
v8	clean split + direct question→option cosine	0.74812	+0.00124
v9	multi-encoder, seed averaging, 2000-row weight fit	0.75602	+0.00790
v10	artifact discriminator, instruction prefixes, weighted blend	0.76142	+0.00540

Phase 1 delivered +0.002 across six submissions. Phase 2 delivered +0.0145 across three — after deleting roughly three-quarters of the code.

6.1 Per-model metrics (Phase 1 system, measured)

Model	MAP@3	Accuracy	Macro F1	Weighted F1
TF-IDF	0.9890	0.9828	0.9824	0.9828
Sentence Transformer	0.9971	0.9951	0.9951	0.9951
RAG	0.9939	0.9902	0.9903	0.9902
MLP (from scratch)	0.9967	0.9951	0.9951	0.9951
LLM (Qwen2.5-14B)	0.9665	0.9363	0.9330	0.9359
NLI cross-encoder	0.6062	0.4436	0.4420	0.4421
Artifact discriminator (OOF)	0.6442	0.5100	0.5082	0.5091
DeBERTa MCQ (OOF)	0.4062	0.2485	0.2378	0.2435
Gated ensemble	0.9988	0.9975	0.9979	0.9975

Rows above 0.95 are inflated by the leak and do not reflect generalisation. The two **out-of-fold** rows are honest, and they are the two that mattered: the discriminator became a core component of the final system, and DeBERTa was removed.

Macro F1 earns its place. For DeBERTa it falls *below* accuracy (0.2378 vs 0.2485) — the signature of a classifier collapsing onto a subset of classes, which neither MAP@3 nor accuracy exposes. With a label distribution ranging from 16.2% to 24.5%, that distinction matters.

7. Error Analysis

Failure 1 — leakage disguised as success. The structural ranker scored a perfect **1.0000** on validation. A model consuming only lengths, positions and similarity statistics cannot solve science questions perfectly. That score was a bug report, and it was recorded as a result.

Failure 2 — a trained model that learned nothing, undetected for six submissions. A loss pinned at exactly `ln(k)` is a one-line assertion away from being caught. It was instead diagnosed only after a full 164-minute run, by which point it had also been consuming 75% of runtime and dragging the ensemble down.

Failure 3 — retrieval-augmented prompting inherited the split's flaws. The 14B model's few-shot examples were the most similar *training* questions **with their correct letters attached**. Because validation $\approx$ training, the retrieved "example" was frequently the same question with the answer supplied:

```
retrieved shot: "What is X?" A).. B).. -> Answer: B      <- same question
target:         "What is X?" A).. B).. -> Answer: ?      <- answer already shown
```

Its 0.9665 / 93.6% top-1 is therefore an upper bound, not a capability estimate — a contaminated split leaks into a zero-shot component that never sees a gradient.

Failure 4 — tuning on too little data. Ensemble weights were fitted on the 408-row validation slice. The standard error of a MAP@3 estimate is $\approx 0.35/\sqrt{n}$: **± 0.018 at $n=408$ versus ± 0.008 at $n=2000$** . At 408 rows the search could not distinguish two weightings differing by less than the entire remaining gap to the cut-off. Refitting on all 2,000 out-of-fold rows was part of the single largest gain (+0.0079).

What generalised. The discriminator, trained 5-fold with `GroupKFold` on cleaned prompts and using no retrieval, was consistent across every fold:

Fold	1	2	3	4	5	OOF
MAP@3	0.6042	0.7121	0.5971	0.6404	0.6675	0.6442

51.0% top-1 against 20% chance, **+0.22 over the trivial baseline**, stable across folds. Distractors in this dataset carry genuinely exploitable artifacts.

8. Insights

- The validation protocol, not the model, was the binding constraint.** Six submissions of architecture work moved the leaderboard +0.002 while validation read 0.99+. Effort spent on the split at the start would have been worth more than every model added afterwards.
- Order preprocessing before splitting.** The leak was a two-line ordering bug: split on raw text, normalise after. Splitting on the post-processed key removed it entirely and was the change that unlocked everything else.
- Always print a trivial baseline beside every result.** `B C A` scores 0.4213. Without it on screen, a 435M model scoring 0.4062 reads as "roughly working".
- Watch the loss value, not just its trend.** A loss sitting at `ln(k)` means no learning has occurred. That assertion costs one line and would have saved two hours per run across six runs.
- A perfect validation score is a bug report.** 1.0000 from a structural-features model should have halted the pipeline.
- Prefer signals whose validation and test performance measure the same thing.** Direct question→option cosine has no corpus and therefore no distribution gap. That single property was worth more than a 14B parameter model.
- Diversity beats capacity in an ensemble.** The discriminator, a 16-feature network, contributed more than Qwen2.5-14B because it was the only component not doing semantic matching.
- Deleting code raised the score.** The final system has ~25 cells against ~60, drops the two most expensive models, and scores 0.0145 higher.

9. Conclusion

The final system fuses seven models — four sentence-transformer encoders compared directly against each option, two supervised neural rankers over pair features, a from-scratch BiLSTM with attention, a TF-IDF baseline, and an answer-artifact discriminator — through calibrated score fusion with weights fitted on 2,000 out-of-fold rows. It scores **0.76142** against a 0.4213 trivial baseline and a 0.73 cut-off.

The more valuable outcome is methodological. A one-step ordering error between splitting and preprocessing produced a validation set that was 99.8% memorisation, which made every model-selection decision unreliable and hid a completely untrained 435M-parameter model inside the ensemble at full weight for six submissions. The measurements that eventually exposed it — a trivial baseline, per-subset metrics, macro F1, out-of-fold estimates, and a loss-value assertion — cost almost nothing to compute. **Building them first would have been worth more than every model built afterwards.**